



IPv6 for Leaders

A Strategic Guide for Decision Makers

Workshop Agenda (1)

Session One (11:45 AM - 1:15 PM)

- IPv6 Basics in Business Terms
- Business Case for IPv6 Adoption
- Security Considerations
- Tracking Adoption with APNIC Labs
- IPv6 and Sustainable Development Goals
- Multi-Stakeholder Model and Governance



Workshop Agenda (2)

Session Two (2:00 PM - 3:30 PM)

- Adoption Barriers and Solutions
- IPv6 Address Planning
- Deployment Planning
- Measuring Success and ROI
- Future-Proofing
- Learning & Development Strategy

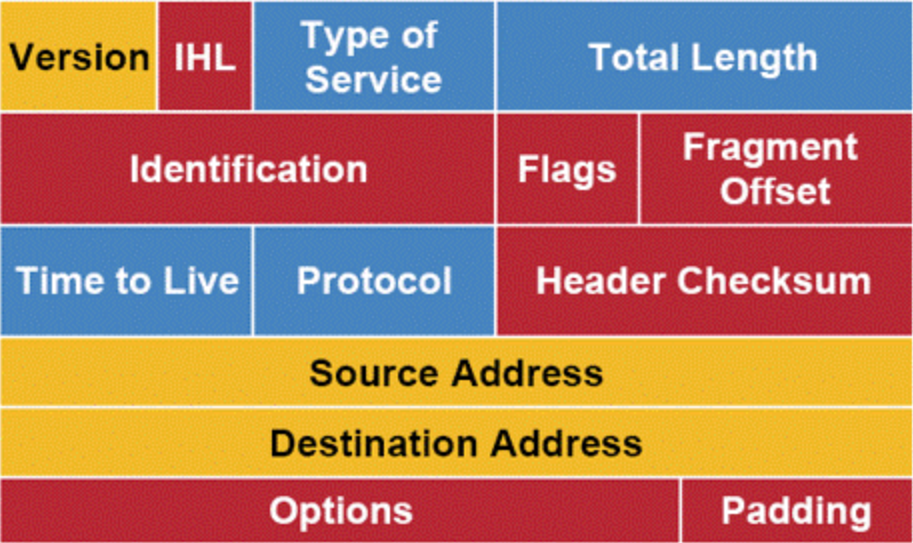


Section 1: IPv6 Basics in Business Terms

What is IPv6?

- Next generation Internet Protocol
- Designed to replace IPv4
- 128-bit addressing (vs. 32-bit for IPv4)
- Enables the continued growth of the Internet
- Foundation for next-generation applications and services

IPv4 Header



Legend

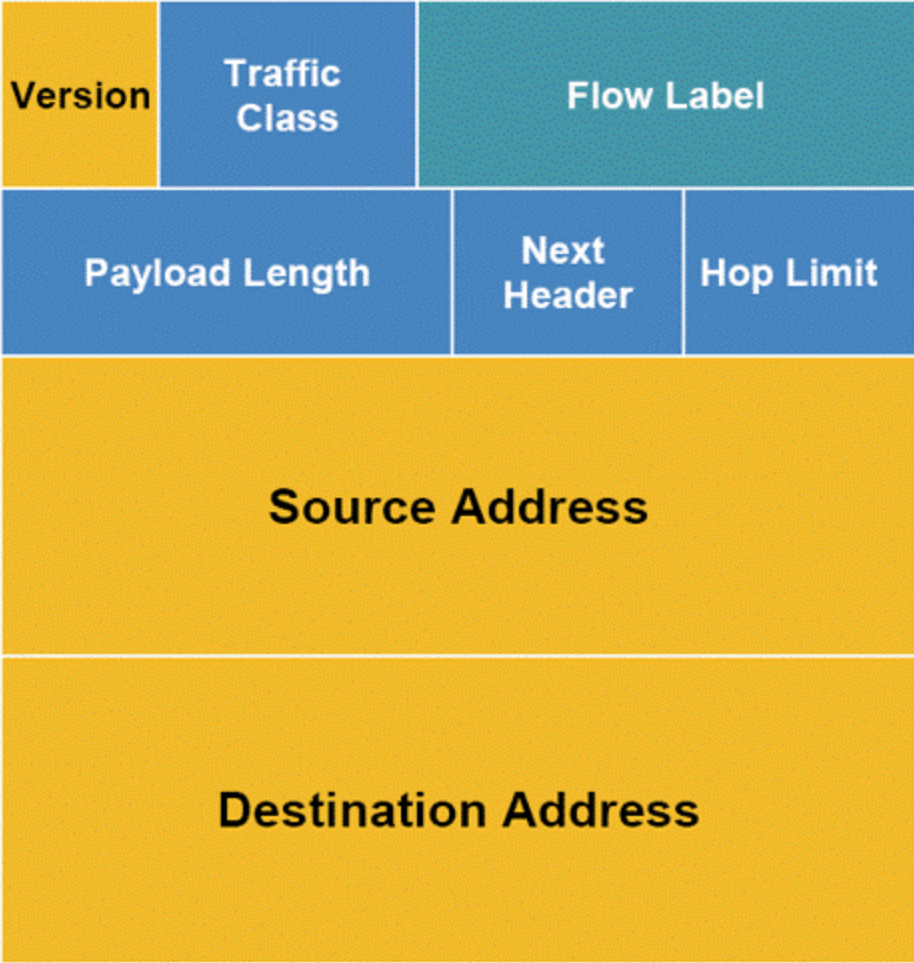
Field's Name Kept from IPv4 to IPv6

Fields Not Kept in IPv6

Name and Position Changed in IPv6

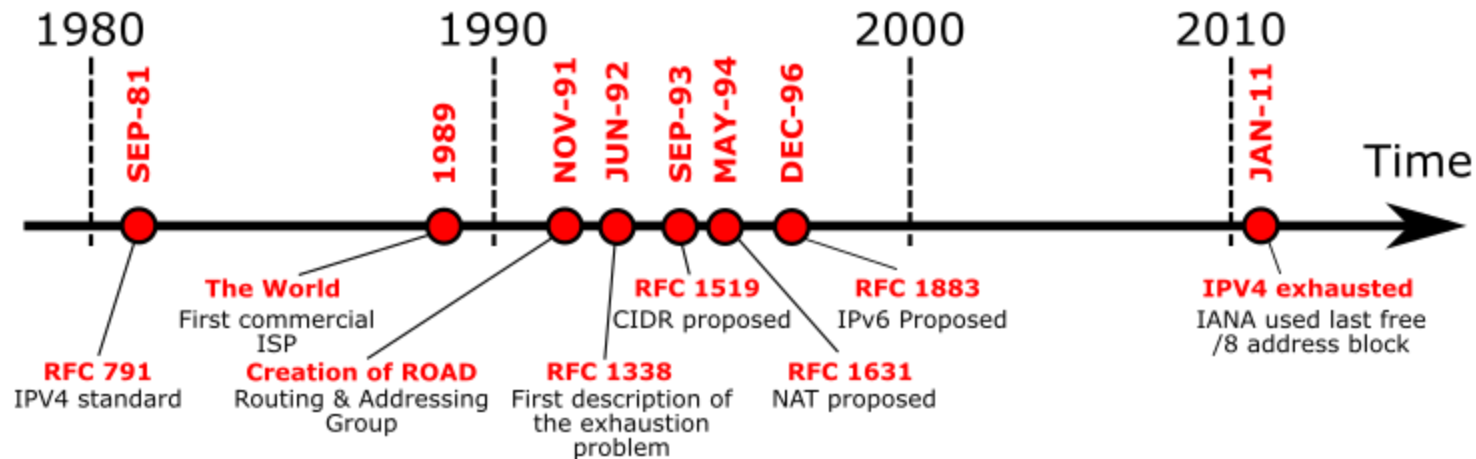
New Field in IPv6

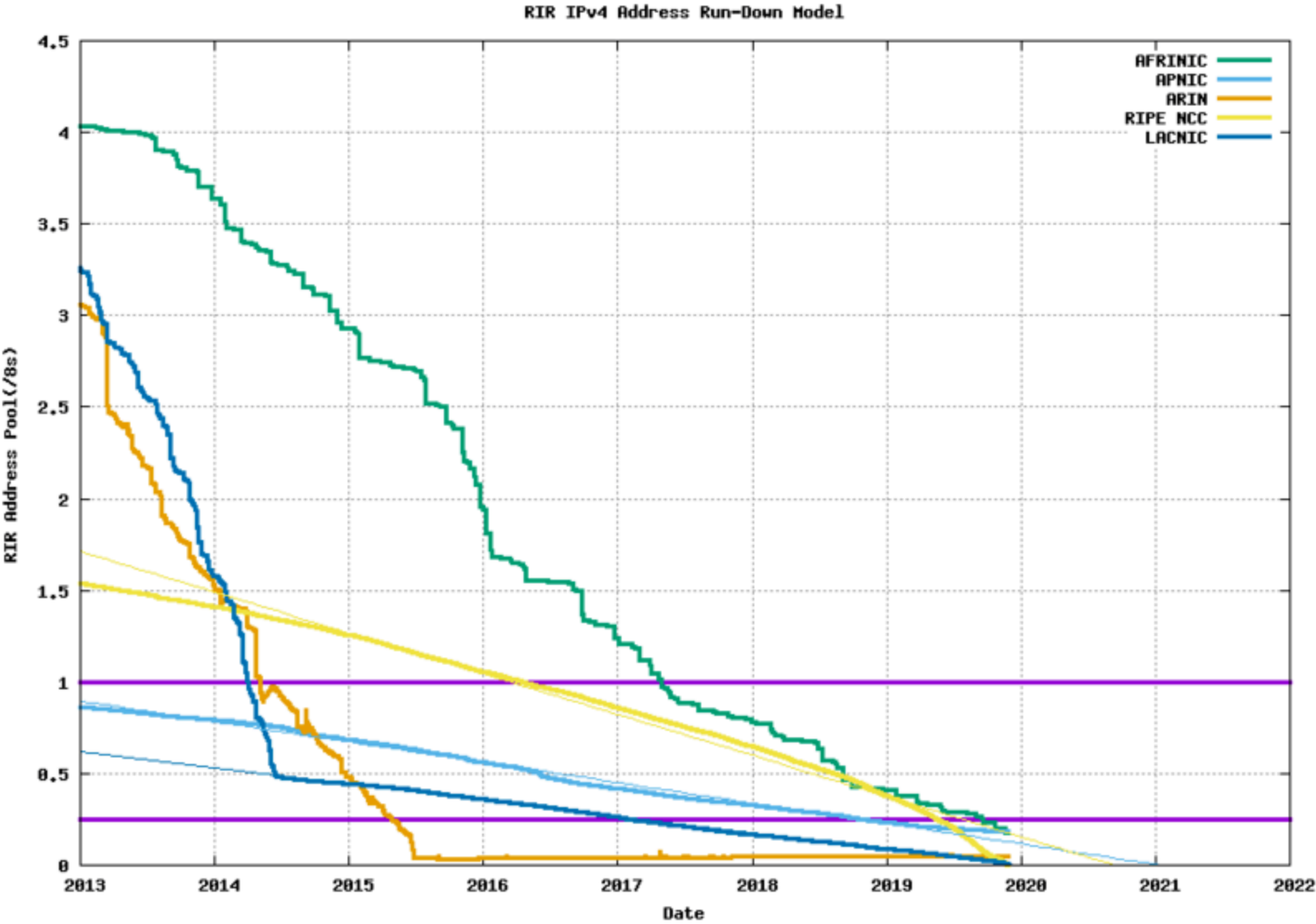
IPv6 Header



IPv4 Address Exhaustion

- IANA global pool exhausted: Early 2011
- Regional registries (RIRs) at various stages of depletion
- Address markets emerging with rising costs
- Increasing use of workarounds (CGNATs, address sharing)
- Business impact: increased costs, reduced functionality





Key Differences That Matter to Leaders

- Address space: 340 undecillion vs. 4.3 billion addresses
- Built-in security with IPsec
- Simplified header structure for better routing efficiency
- Auto-configuration capabilities
- Better support for mobile networks

The Scarcity to Abundance Mindset

- IPv4: Conservation mindset
 - Careful address allocation
 - NAT as standard practice
 - IP addresses as limited resource
- IPv6: Abundance mindset
 - Generous subnet allocation
 - Direct end-to-end connectivity
 - IP addresses as unlimited resource

Section 2: Business Case for IPv6 Adoption

Future-Proofing Your Digital Infrastructure

- Preparing for Internet of Things expansion
- Supporting cloud-native architectures
- Enabling edge computing models
- Simplification of network management
- Elimination of NAT complexity
- Direct addressing for security and monitoring

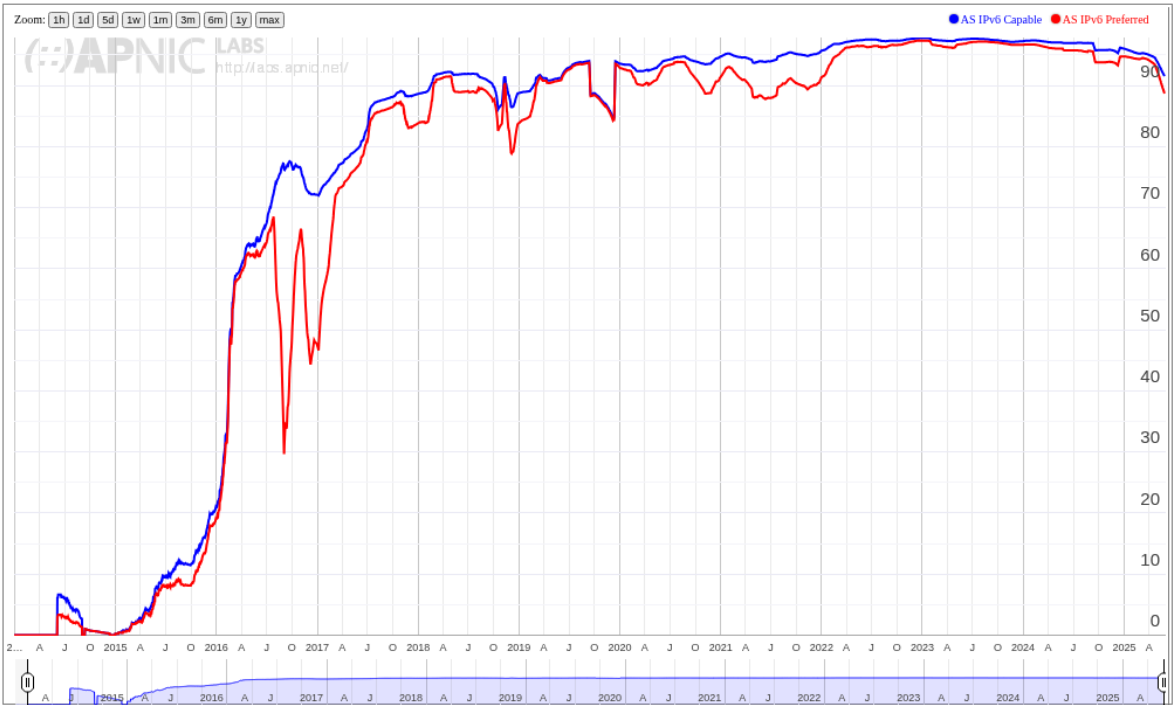
Competitive Advantages of Early Adoption

- Market differentiation opportunities
- Avoiding last-minute rushed implementations
- Lower long-term costs with planned transitions
- Enhanced capability for innovative services
- Improved customer experience (especially mobile)
- Talent attraction and retention advantages

The Growth Problem

- IPv4 scarcity as barrier to internet growth
- Rising costs of IPv4 addresses (\$50-60 per address)
- Service degradation from address sharing
- Impact on developing regions and new entrants
- Connection between IPv6 and digital inclusion

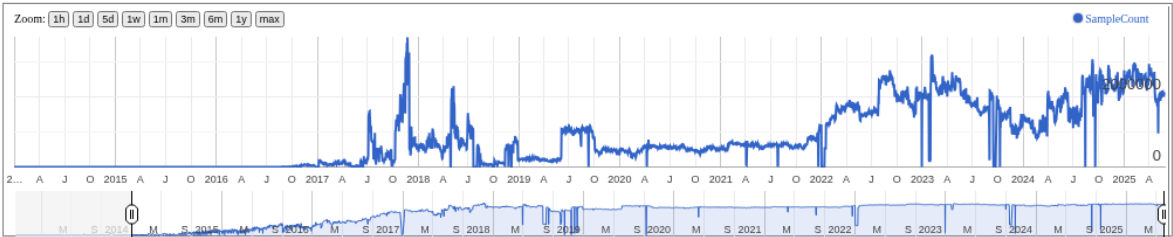
IPv6 Per-Country Deployment for AS55836: RELIANCEJIO-IN Reliance Jio Infocomm Limited, India (IN)



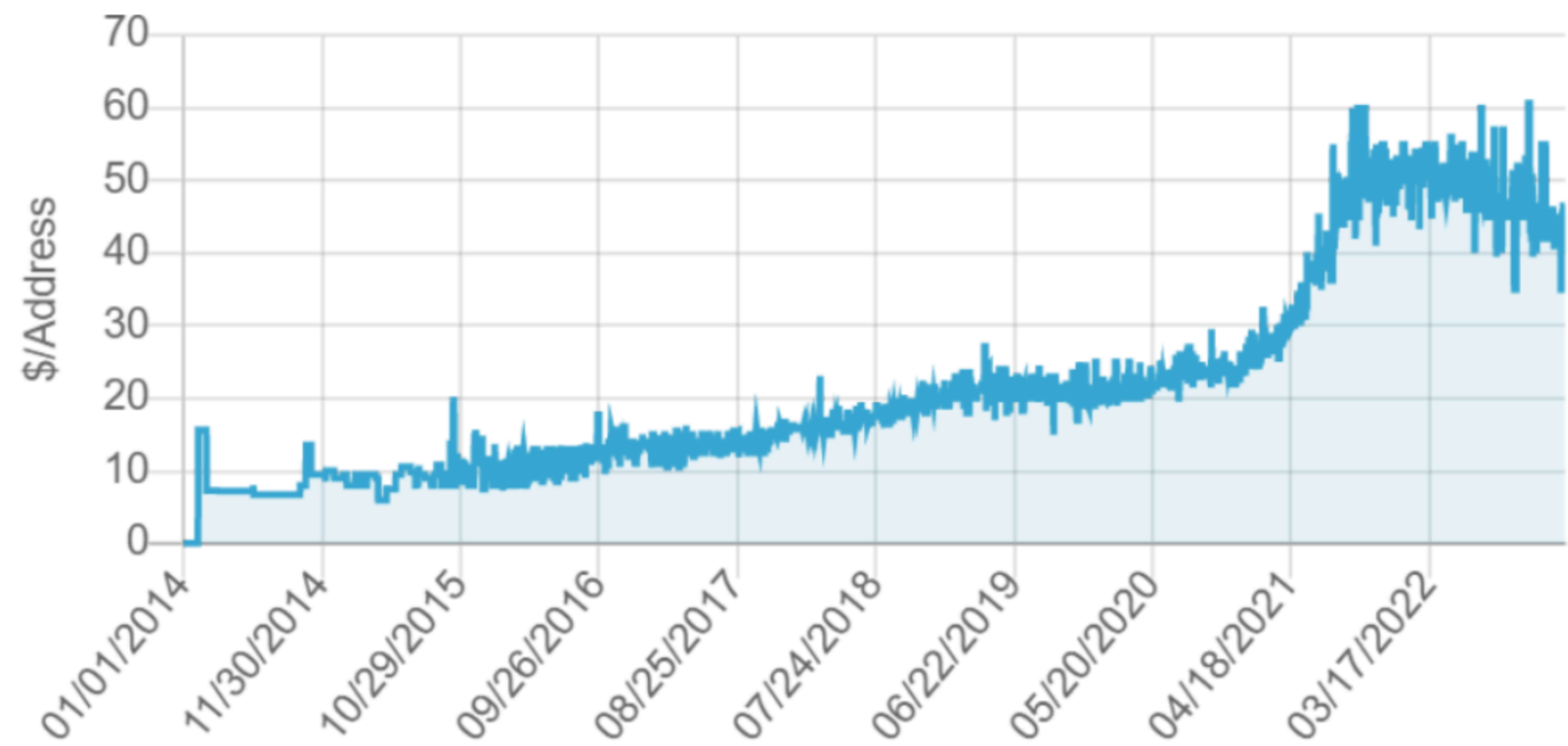
Average Interval (days) 90

- ☐ Show country ASN List
 - ☐ Hide IPv6 Preferred
 - ☐ Show Country Totals
- Redraw

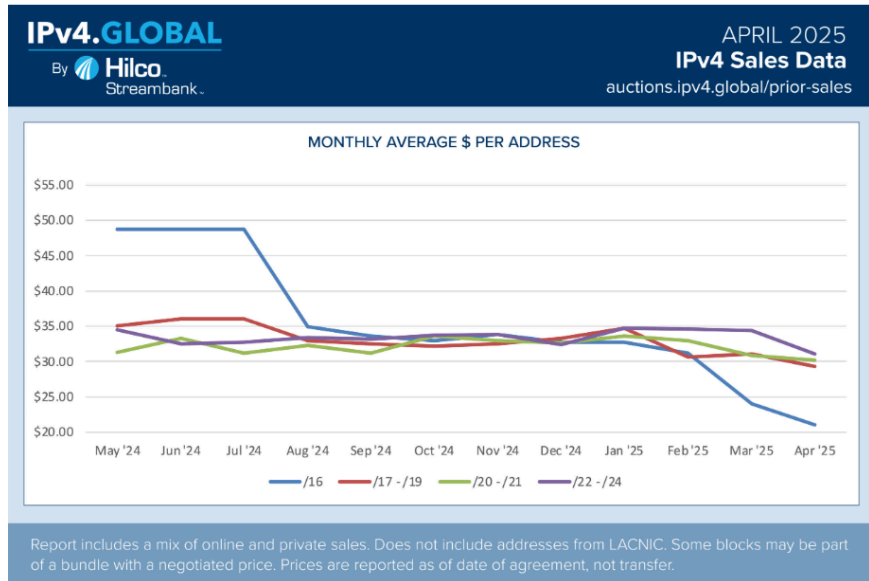
[Download JSON Time Series Data for AS55836 in IN](#)



The Cost Of IPv4 Addresses



April 2025 IPv4 Marketplace Sales Report



IPv4 prices continue to inch down, while volume remains steady. The gap between large and small blocks continues to grow as well. Significant supply flowing into the market has contributed to the price drop. As that surplus works through the market, prices may not continue to decline.

Prices are volatile

Prices are trending downwards

Prices are varied by size

WHY?

Section 3: Security Considerations in IPv6 Transitions

Security Myths and Realities

Myth: IPv6 is inherently more secure than IPv4

Reality: IPv6 has different security characteristics, not inherently better

Myth: IPv6 eliminates the need for firewalls

Reality: Firewalls remain essential in IPv6 deployments

Myth: NAT provides security

Reality: NAT provides obscurity, not true security

Security Constants Across Protocols

- Defense in depth remains essential
- User authentication requirements unchanged
- Application security equally important
- Data encryption still necessary
- Monitoring and logging critical

Beyond NAT: Security in an IPv6 World

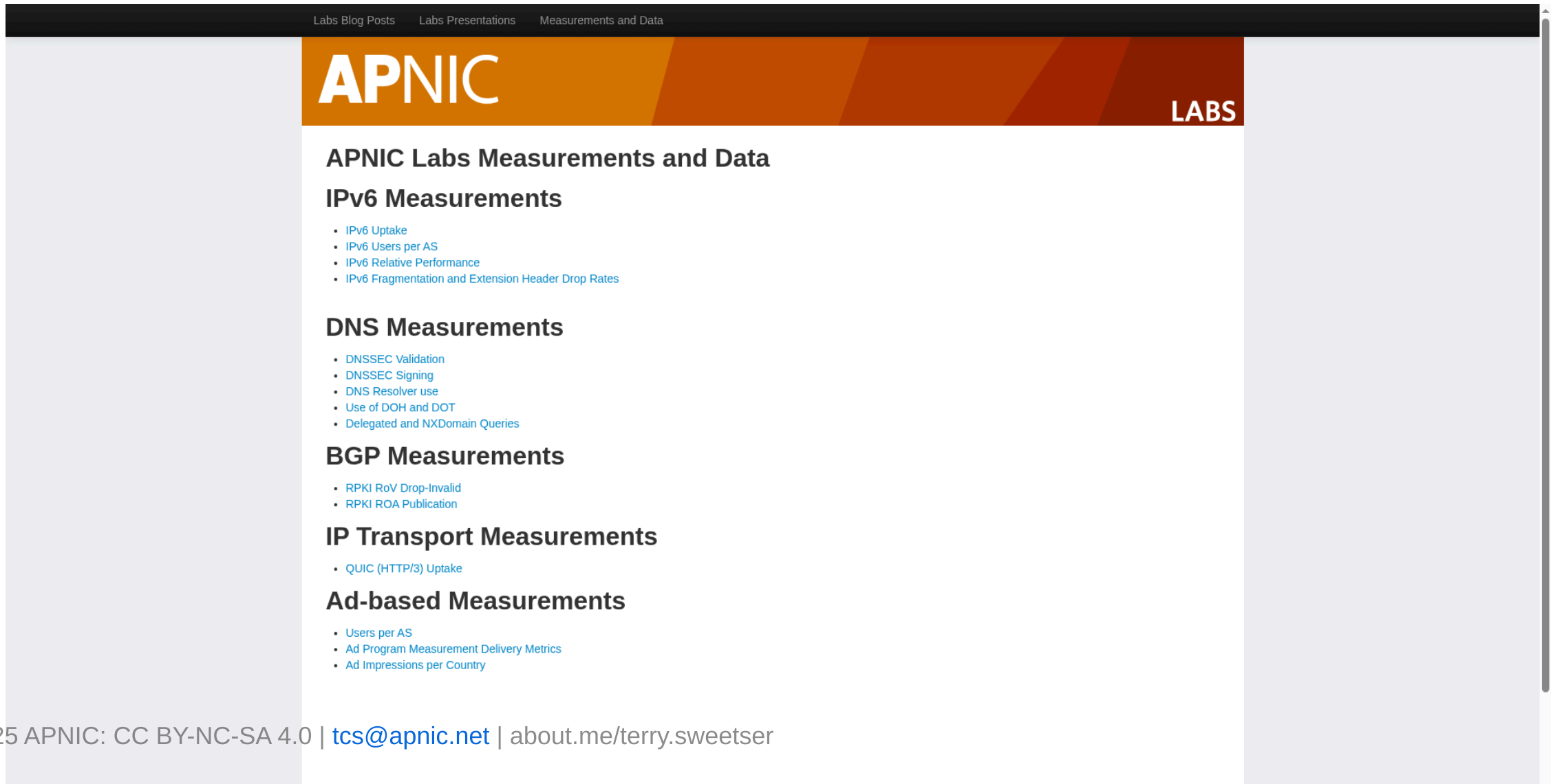
- Direct addressing doesn't mean direct access
- Stateful and stateless firewalls remain essential
- Address plan segmentation for security zones
- Unique security considerations:
 - Larger scanning space
 - Extension header inspection
 - Neighbor Discovery Protocol protection

IPv6 Security Governance

- Security policy updates for IPv6
- Risk assessment for transition period
- Dual-stack security considerations
- Skills development for security teams
- Monitoring strategy adjustments
- Incident response procedure updates

Section 4: Tracking Adoption with APNIC Labs

<https://labs.apnic.net/measurements/>



Labs Blog Posts Labs Presentations Measurements and Data

APNIC

LABS

APNIC Labs Measurements and Data

IPv6 Measurements

- [IPv6 Uptake](#)
- [IPv6 Users per AS](#)
- [IPv6 Relative Performance](#)
- [IPv6 Fragmentation and Extension Header Drop Rates](#)

DNS Measurements

- [DNSSEC Validation](#)
- [DNSSEC Signing](#)
- [DNS Resolver use](#)
- [Use of DOH and DOT](#)
- [Delegated and NXDomain Queries](#)

BGP Measurements

- [RPKI RoV Drop-Invalid](#)
- [RPKI ROA Publication](#)

IP Transport Measurements

- [QUIC \(HTTP/3\) Uptake](#)

Ad-based Measurements

- [Users per AS](#)
- [Ad Program Measurement Delivery Metrics](#)
- [Ad Impressions per Country](#)

Global and Regional Adoption Trends

- Current global IPv6 adoption: ~40% (varies by measurement method)
- Leading countries: India, Belgium, Germany, USA
- Leading regions: South Asia, Europe, North America
- Mobile networks driving much of the adoption
- Wide variation in enterprise adoption

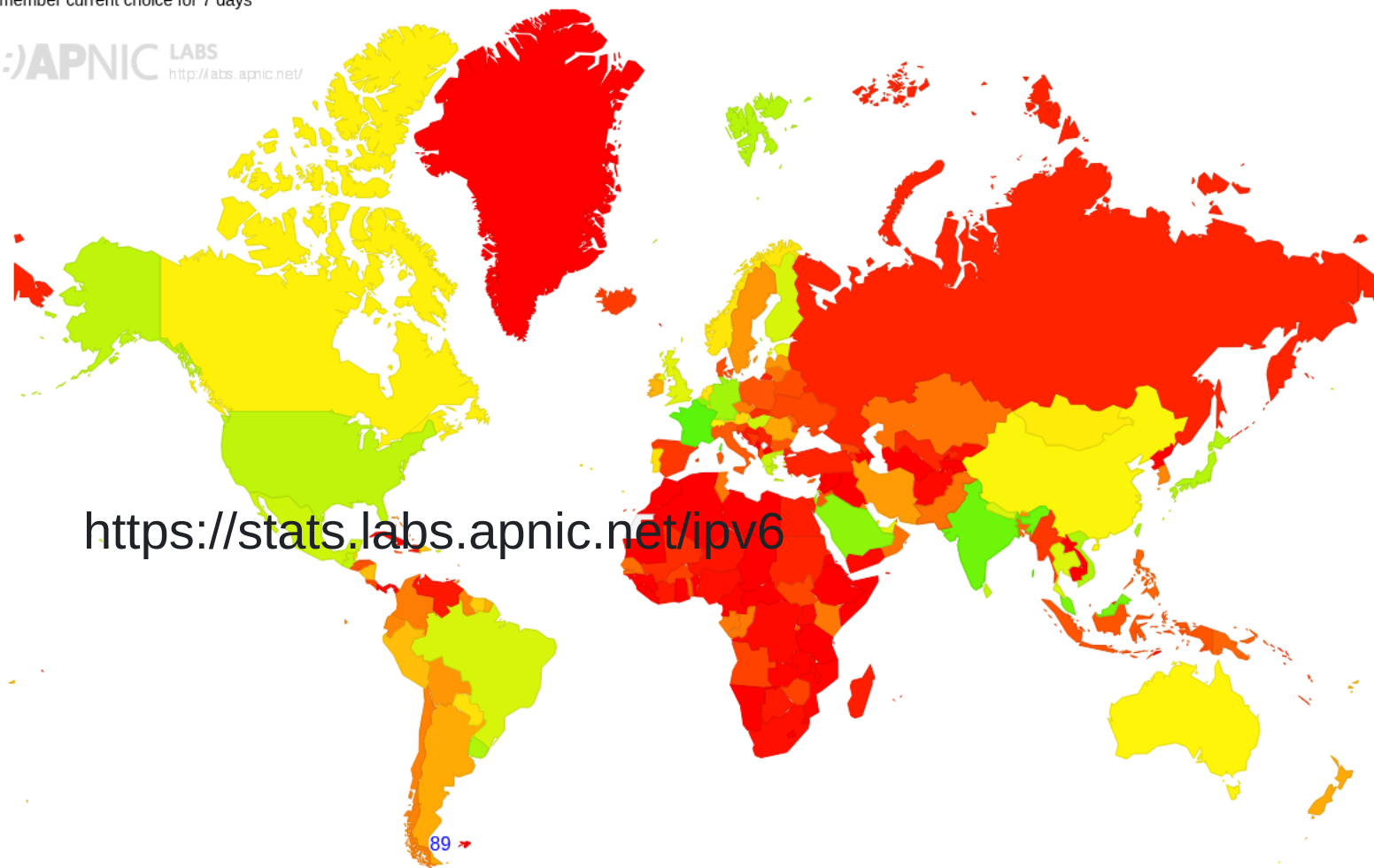
IPv6 Capable Rate by country (%)

IPv6 for Leaders Workshop

[Click here for a zoomable map](#)

☐ Remember current choice for 7 days

(::)APNIC LABS
<http://labs.apnic.net/>



<https://stats.labs.apnic.net/ipv6>

Use of IPv6 for World (XA)



<https://stats.labs.apnic.net/ipv6>

Use of IPv6 for Polynesia (QS)



<https://stats.labs.apnic.net/ipv6/QS>

Using APNIC Labs Statistics

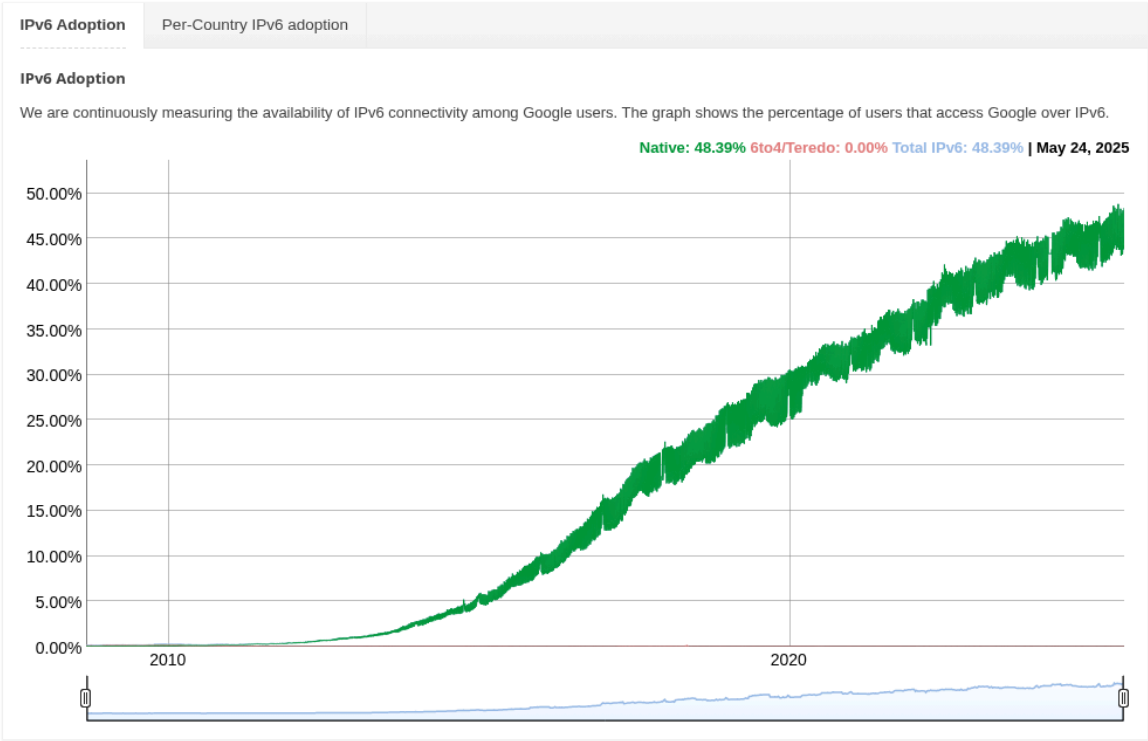
- Methodology: browser-based measurement
- Country and network-level statistics
- Historical trends and growth patterns
- Different measurement types:
 - Preferred address selection
 - DNS resolution capabilities

<https://www.google.com/intl/en/ipv6/statistics.html>

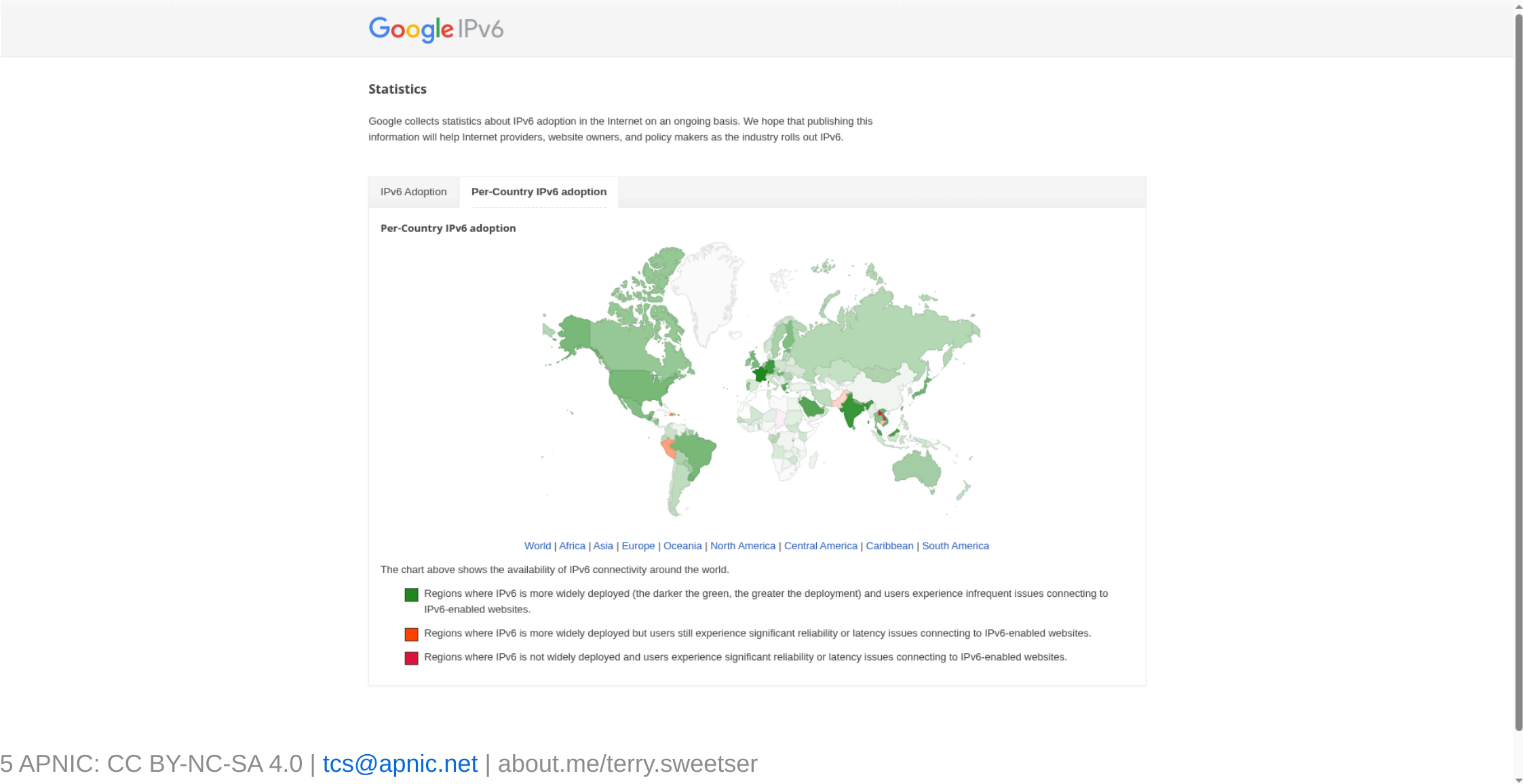


Statistics

Google collects statistics about IPv6 adoption in the Internet on an ongoing basis. We hope that publishing this information will help Internet providers, website owners, and policy makers as the industry rolls out IPv6.



<https://www.google.com/intl/en/ipv6/statistics.html>



Benchmarking Your Organization

- Comparing to industry peers
- Geographic considerations
- Identifying adoption gaps
- Setting realistic targets
- Tracking progress over time
- Using data to inform strategy

Section 5: IPv6 and Sustainable Development Goals

IPv6 and SDG Alignment

- SDG 9: Industry, Innovation and Infrastructure
 - Expanded connectivity
- SDG 11: Sustainable Cities and Communities
 - Smart city applications
- SDG 10: Reduced Inequalities
 - Bridging digital divide
- SDG 13: Climate Action
 - More efficient networks

Case Studies: IPv6 and Sustainability

Case Study 1: Rural Connectivity

- Enhanced Network Management and Performance
- Energy Efficiency and Sustainability
- Future-Proofing Rural Networks

Case Study 2: Smart City Implementation

- Cost-Effective Deployment and Operation
- Better Support for Growing Number of Devices

<https://www.mdpi.com/605136>

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Article

Affordable Broadband with Software Defined IPv6 Network for Developing Rural Communities

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(This article belongs to the Section **Information Systems**)

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Abstract

The software defined networking (SDN) paradigm with enhanced features of IPv6 offers flexible network

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The realities of building an IPv6-only city

By [Guoliang Yang](#) on 29 Oct 2024

Category: [Tech matters](#)

Tags: [APNIC 58](#), [China](#), [Guest Post](#), [IPv6](#)

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Xiong'an New Area.

As previously discussed [in this post](#), Xiong'an New Area (Xiong'an), a pilot city established in 2017 about 100 kilometres west of Beijing, aims to be a model for future digital cities, built with IPv6-only infrastructure from the start. To achieve this, the local government is prioritizing IPv6 in its top-level planning, designing for the deployment of 198,000 sensors per square kilometre, and supporting over 1 million Internet of Things (IoT) devices per square kilometre. This post will look at the lessons the Xiong'an team has learned so far from building this IPv6-only city.

IPv6-only or IPv4 and IPv6 dual-stack?

As a new city, how should Xiong'an approach IPv6 deployment — IPv6-only or IPv4 and IPv6 dual-stack? While IPv6-only, as shown in Figure 1, simplifies network, application, and terminal device management, compatibility issues make it challenging. On the other hand, dual-stack is easier to implement

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Section 6: Multi-Stakeholder Model and IPv6 Governance

Internet Governance Framework

- Multi-stakeholder model vs. traditional approaches
- Key principles:
 - Bottom-up coordination
 - Transparency
 - Consensus-based decision making
 - Technical community leadership
 - Cross-sector participation

Key Players in IPv6 Governance

- Internet Corporation for Assigned Names and Numbers (ICANN)
- Internet Engineering Task Force (IETF)
- Regional Internet Registries (RIRs)
 - APNIC, ARIN, RIPE NCC, LACNIC, AFRINIC
- National IPv6 task forces
- Network operator groups

Interactive Discussion #1 -- IPv6 Readiness Assessment

1. Form small groups (4-5 people)
2. Complete the assessment worksheet:
 - Technical infrastructure
 - Staff knowledge and skills
 - Business case understanding
 - Executive support
 - Security preparedness
3. Identify top 3 strengths and 3 gaps
4. Share key insights (2-3 minutes per group)

Section 7: Common Adoption Barriers and Solutions

Technical and Knowledge Barriers

Barriers	Solutions
Limited IPv6 expertise	Targeted training programs
Incomplete vendor support	Vendor assessment
Integration challenges	Phased implementation
Perception of complexity	Proof of concept deployments

Financial Considerations

Barriers	Solutions
Unclear ROI	Business case frameworks
Competition for IT resources	Integration with refresh cycles
Legacy system replacement costs	Cost avoidance quantification
Operational expense concerns	Staged investment approach

Organizational Change Management

Barriers	Solutions
Resistance ("IPv4 works fine")	Clear articulation of benefits
Competing priorities	Strategic alignment
Lack of executive sponsorship	Executive education
Siloed responsibility	Cross-functional teams

Section 8: What to Look for in an IPv6 Address Plan

Strategic Allocation Principles

- Hierarchical design for scalability
- Allocation strategy by:
 - Geographic location
 - Business function
 - Security zones
 - Growth accommodation
- Numbering conventions
- Readability vs. efficiency

Security by Design in Address Planning

- Isolation of security domains
- Unpredictable addressing where appropriate
- Stable addressing for critical infrastructure
- Transition mechanism addressing
- Temporary address considerations
- Alignment with security policy requirements

Organizational Structure in Addressing

- Address delegation strategy
- Department/function reflection in addressing
- Site/location considerations
- Service-based addressing
- Growth accommodations
- Documentation and governance

Section 9: Evaluating a Deployment Plan



THE FIRST STEPS IN IPV6 ADOPTION – HAVING A PLAN



October 28, 2014



Title: The first steps in IPv6 adoption – having a plan



If you are like the majority of IT professionals who are operating networks and maintaining systems today you likely haven't implemented IPv6 at all. Most companies have not had the time, money nor need to get IPv6 deployed and have stayed on IPv4. Therefore IPv6 adoption is further behind than many other technologies that were developed along a comparable time line. Now that IPv6 is rising rapidly as a technology people should pay attention to and know (for a variety of reasons: see some previous posts [here](#), [here](#) and [here](#)), what can you do to get started? What are the critical things you need to know?

To begin with, IPv6 is no different than any other project you might have as an IT professional (though perhaps a bit bigger in scope and impact). Like all projects it requires a plan to lay



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Internet Engineering Task Force (IETF)
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Category: Informational
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October 2014

Enterprise IPv6 Deployment Guidelines

Abstract

Enterprise network administrators worldwide are in various stages of preparing for or deploying IPv6 into their networks. The administrators face different challenges than operators of Internet access providers and have reasons for different priorities. The overall problem for many administrators will be to offer Internet-facing services over IPv6 while continuing to support IPv4, and while introducing IPv6 access within the enterprise IT network. The overall transition will take most networks from an IPv4-only environment to a dual-stack network environment and eventually an IPv6-only operating mode. This document helps provide a framework for enterprise network architects or administrators who may be faced with many of these challenges as they consider their IPv6 support strategies.

Status of This Memo

This document is not an Internet Standards Track specification; it is published for informational purposes.

This document is a product of the Internet Engineering Task Force (IETF). It represents the consensus of the IETF community. It has received public review and has been approved for publication by the Internet Engineering Steering Group (IESG). Not all documents approved by the IESG are a candidate for any level of Internet Standard; see [Section 2 of RFC 5741](#).

Information about the current status of this document, any errata, and how to provide feedback on it may be obtained at <http://www.rfc-editor.org/info/rfc7381>.

Datatracker

RFC 7381

RFC - Informational

Info

Contents

Prefs

Document type

RFC - Informational

October 2014

Report errata

Was [draft-ietf-v6ops-enterprise-incremental-ipv6](#) (v6ops WG)

Select version

00

01

02

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04

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06

RFC 7381

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draft-ietf-v6ops-enterprise-incremental-ipv6-06

RFC 7381

Side-by-side

Inline

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RFC stream

Other formats

txt

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Additional resources

[Mailing list discussion](#)

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Milestone Planning for IPv6

- Assessment and inventory phase
- Address planning phase
- Core infrastructure enablement
- Security implementation
- Application testing
- Pilot deployments
- Production deployment
- Monitoring and optimization

Risk Management Approaches

Key risks to address	Risk mitigation strategies
Service disruption during transition	Comprehensive testing
Security vulnerabilities	Phased deployment
Application compatibility	Rollback capabilities
Performance issues	Monitoring and alerting
Staff readiness	Transition mechanism selection

Transition Mechanisms

- Dual-stack implementation
- Tunneling approaches:
 - 6to4, 6in4, 6rd
 - DS-Lite, MAP-T/MAP-E
- Translation mechanisms:
 - NAT64/DNS64
- Selection criteria:
 - Environment constraints
 - Performance requirements
 - Support and security

Section 10: Measuring Success and ROI

Key Performance Indicators (1)

Technical KPIs:

- Percentage of IPv6-enabled infrastructure
- IPv6 traffic volume
- Performance metrics
- Incident frequency

Key Performance Indicators (2)

Business KPIs:

- Cost avoidance from IPv4 purchases
- Operational simplification metrics
- Support incident reduction
- New capability enablement

ROI Calculation Methodology

Costs to consider:

- Hardware/software upgrades
- Training and certification
- Consulting services
- Staff time
- Potential disruption

ROI Calculation Methodology

Benefits to quantify:

- IPv4 cost avoidance
- Operational efficiency gains
- Risk reduction value
- New business capabilities
- Competitive positioning

Executive Dashboard Example

- High-level implementation status
- Key milestones achieved
- Risk summary
- Cost tracking
- Business benefits realized
- Next steps and decisions needed

Section 11: Future-Proofing Beyond Basic IPv6 Adoption

IoT and Massive Scaling

- Projected: 75 billion IoT devices by 2030
- Address requirements for direct connectivity
- Sensor network architectures
- Edge processing models
- Security for massive deployments
- Management at scale

Edge Computing and 5G/6G

- Distributed computing enabled by IPv6
- Mobile network evolution requirements
- Low-latency applications
- Network slicing capabilities
- End-to-end connectivity models
- Architecture evolution

Strategic Vision for Connected Operations

- Unified addressing across environments
- Seamless OT/IT integration
- Zero-trust security models
- Data-driven operations
- Customer and partner integration
- New business models

Section 12: Learning & Development Strategy for IPv6

Building a Competency Framework

- Role-based IPv6 knowledge requirements:
 - Network engineers
 - Security specialists
 - Application developers
 - IT operations
 - Project managers
 - Executives
- Skill assessment
- Certification pathways

APNIC and NGO Resources

- APNIC training offerings:
 - In-person workshops
 - Online courses
 - Lab environments
 - Certification programs
- Internet Society resources
- Regional NOG events
- Open educational materials
- Community support

Knowledge Sharing Best Practices

- Internal community of practice
- Lab environments for hands-on learning
- Lunch and learn sessions
- Implementation documentation
- Lessons learned repositories
- Mentoring programs

Interactive Discussion #2

Deployment Roadmap Development

1. Individual work (15 minutes):

- Draft initial 90-day roadmap
- Focus on executive sponsorship
- Identify assessment activities
- Define skills development priorities
- Plan early wins
- Outline resource requirements

2. Small group sharing (10 minutes)

Next Steps and Resources

- APNIC IPv6 Program: <https://www.apnic.net/ipv6>
- Regional IPv6 Task Force contacts
- Workshop materials available online
- Follow-up workshop opportunities
- Implementation support resources
- Community engagement channels

Thank You!

Questions and Discussion